

GMS - Architecture & Implementation Design

1. Introduction

Gmail Management Service is a backend service designed to automate Gmail inbox management. It retrieves emails from the past 24 hours, analyzes those mails and categorizes them, and generates summaries.

The processed emails are combined into a **Single Email - Digest** which contains details such as the **Sender, Subject, and Summary**, along with action buttons such as **Delete, Archive, Mark as Important, Mark as Spam, and Reply**. The digest is automatically delivered to the user at a scheduled time.

2. Goal

The goal of **Gmail Management Service** is to automatically process emails received in the past 24 hours and provide the user with a **single daily email** containing **categorized emails, their summaries, and the required action buttons**.

3. Background / Current State

Email management is generally performed manually by users. Users need to review incoming emails, identify important messages, and decide how each email should be handled.

As the number of emails increases, it becomes difficult to quickly identify relevant messages and keep track of all emails received during the day. Emails may contain different types of information, including important messages, normal communication, promotional content, and spam.

GMS is designed to simplify this process by automatically processing emails received during the previous 24 hours. The service categorizes the emails, generates summaries, and combines the processed information into a **single daily email digest**. The digest also provides actions for managing individual emails.

4. Requirements

The Gmail Management Service should:

- Retrieve emails received within the **past 24 hours**.
- Analyze and categorize emails as **Important, Normal, Promotional, or Spam using AI**.
- Generate a **summary** for each processed email.
- Generate a **single daily email** containing the categorized emails and their summaries.
- Provide actions buttons such as **Delete, Archive, Mark as Important, and Reply**.
- Automatically deliver the digest to the user at a **scheduled time using AWS**.